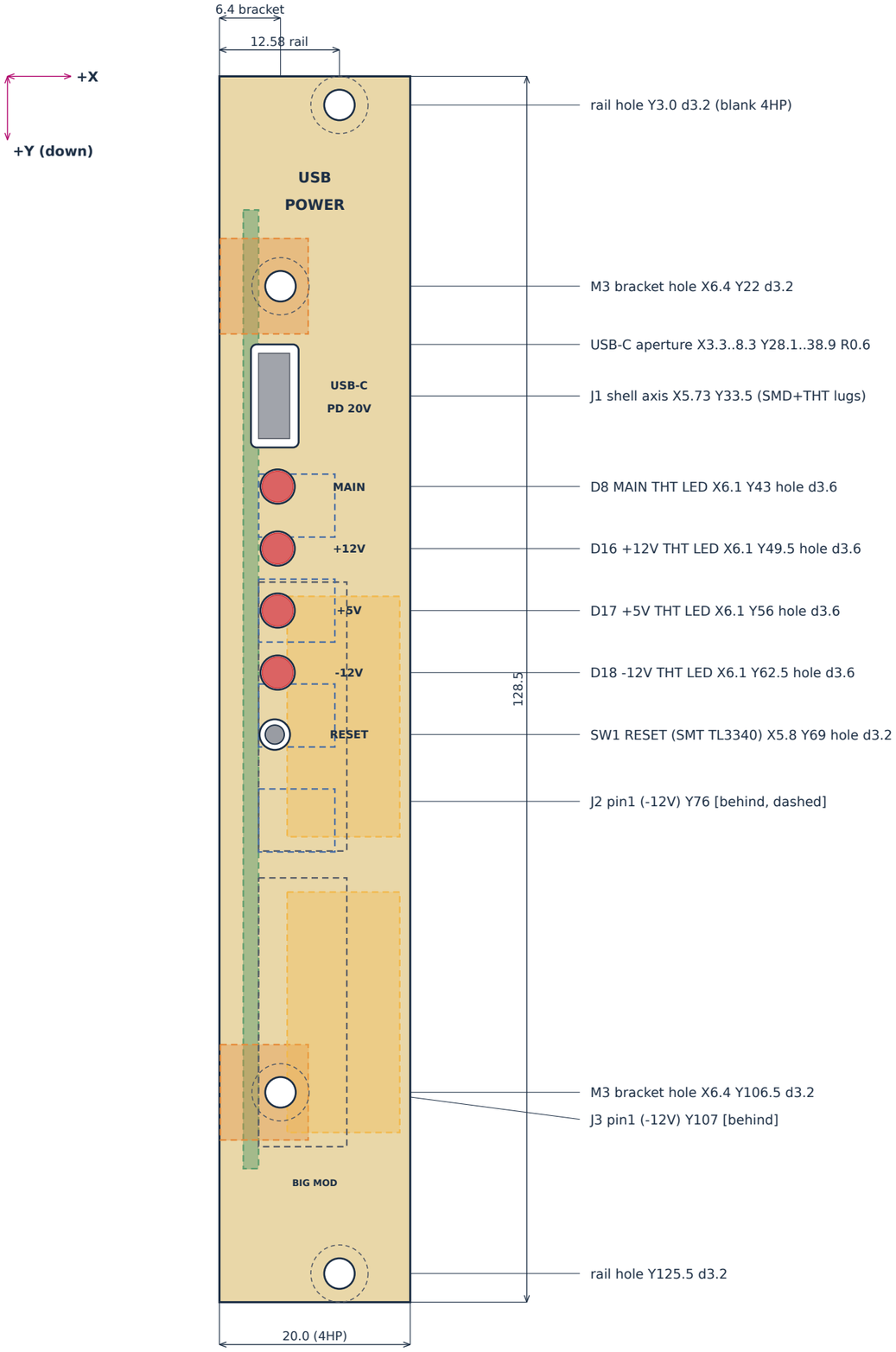
usb-power-45w - FRONT PANEL VIEW (4HP)

Seen from the front; X across, Y down; origin = panel upper-left front corner. Solid = panel features; dashed = parts behind the panel (projected).



Native 4HP panel PCB 20.0 x 128.5 x 1.6 (passive, NPTH only)
Carrier edge behind panel: B.Cu X2.5 / F.Cu X4.1 (copied)
Two plain M3 L brackets (foot X0.1..9.3)
THT 3 mm LEDs D8 MAIN, D16 +12V, D17 +5V, D18 -12V
J1 USB-C body / SW1 actuator (no verified model: envelope)
PROPOSAL envelope: mated IDC plugs on J2/J3 (behind)
PROVISIONAL cans C5/C34/C35/C36 (height context)

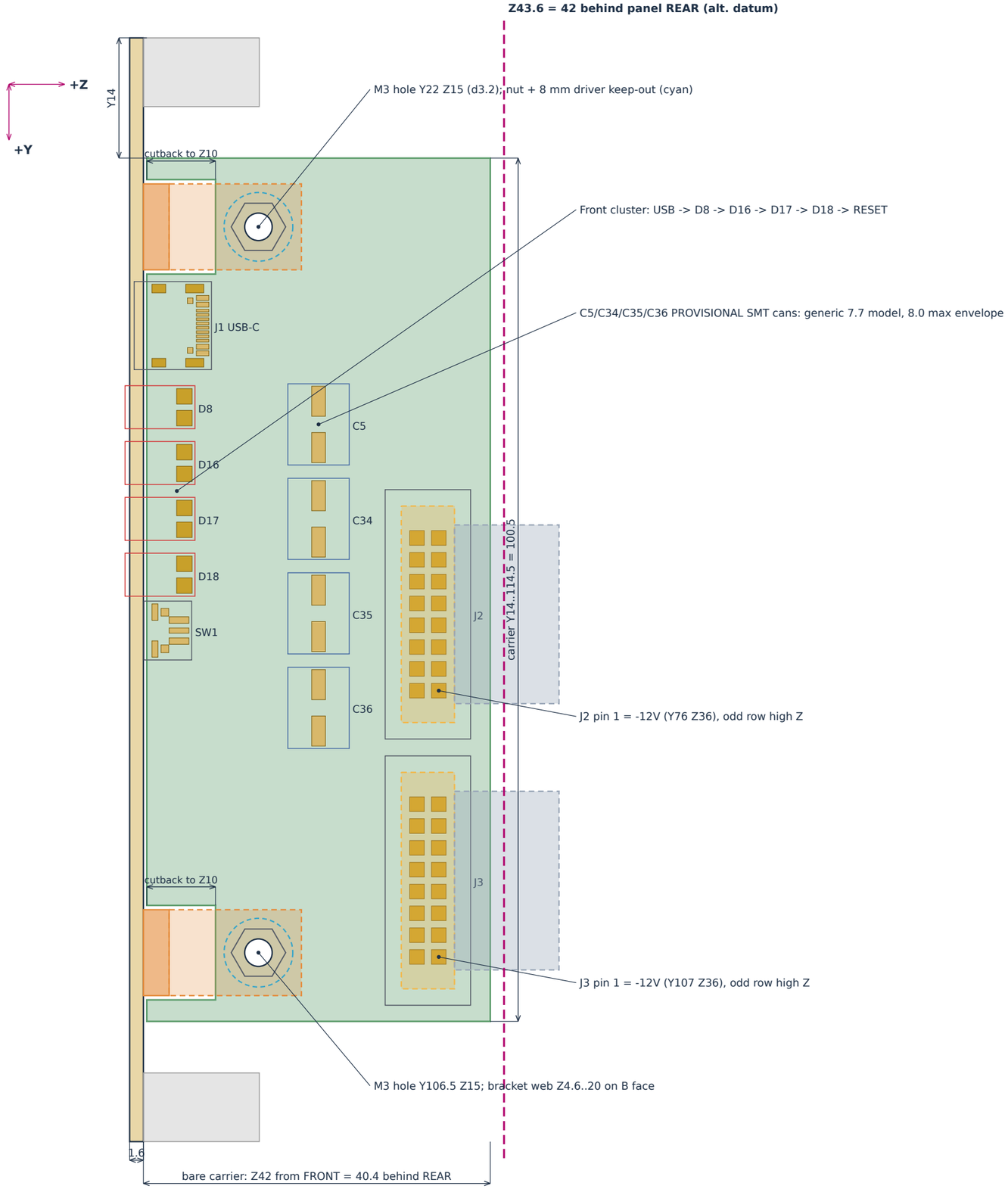
Checks (nominal):

- Panel = blank 4HP precedent: True; rail holes != bracket holes
- LED/SW1 axes on hole centres: dY = dX = 0.000; LED body radial clearance 0.175 mm
- SW1 actuator radial clearance 0.6 mm; USB shell-to-aperture 0.93 (Y) / 0.87 (X) mm/side
- Min panel web between openings 2.3 mm; USB aperture to bracket head 3.1 mm
- Rail screw heads d6 (dotted) clear the cluster; bracket screw heads d6 dotted
- Panel native DRC: 0 violations (research/mech-preview/panel-drc.json)

Copied unchanged from usb-power: aperture, LED X6.1 / hole d3.6, SW1 X5.8 / hole d3.2, USB axis X5.73 Y33.5, bracket holes. Changed: 4HP width, rail X12.58, 4 LEDs, RESET Y69.

usb-power-45w - SIDE VIEW OF THE CARRIER F FACE (YZ, seen from +X)

Z (into case) to the right, Y down. Placed parts: courtyards (thin) + pads (filled). Bracket web is on the hidden B face (dashed). 205 other SMD parts are PARKED OFF-BOARD (not placed).



Carrier (native outline, 4-layer, unrouted)
Bracket foot (solid) / web on B face (dashed)
PROPOSAL mated IDC plug envelope
PROPOSAL ribbon leaving +Z toward the bus
F-side M3 nut driver keep-out (d8)
Case rail section (context)

Checks / findings (not waived):

- Courtyard to edge < 1 mm: D16 -1.535, D17 -1.535, D18 -1.535, D8 -1.535, J1 -0.875, SW1 -0.325
- Pad copper to edge < 1 mm: J1 0.604, SW1 0.6 (DRC errors)

-> edge-mounted panel parts copied from usb-power; USER DECISION needed

- J2/J3 courtyard to edge 2.295/1.875 mm; cans >= 15.303 mm
- Hand-solder 2 mm: closest D18-SW1 2.734 mm, D8-J1 3.85 mm
- Area estimate (NON-BLOCKING): courtyard bbox sum 2833 mm2 vs ~2491 mm2 F (x1.14)

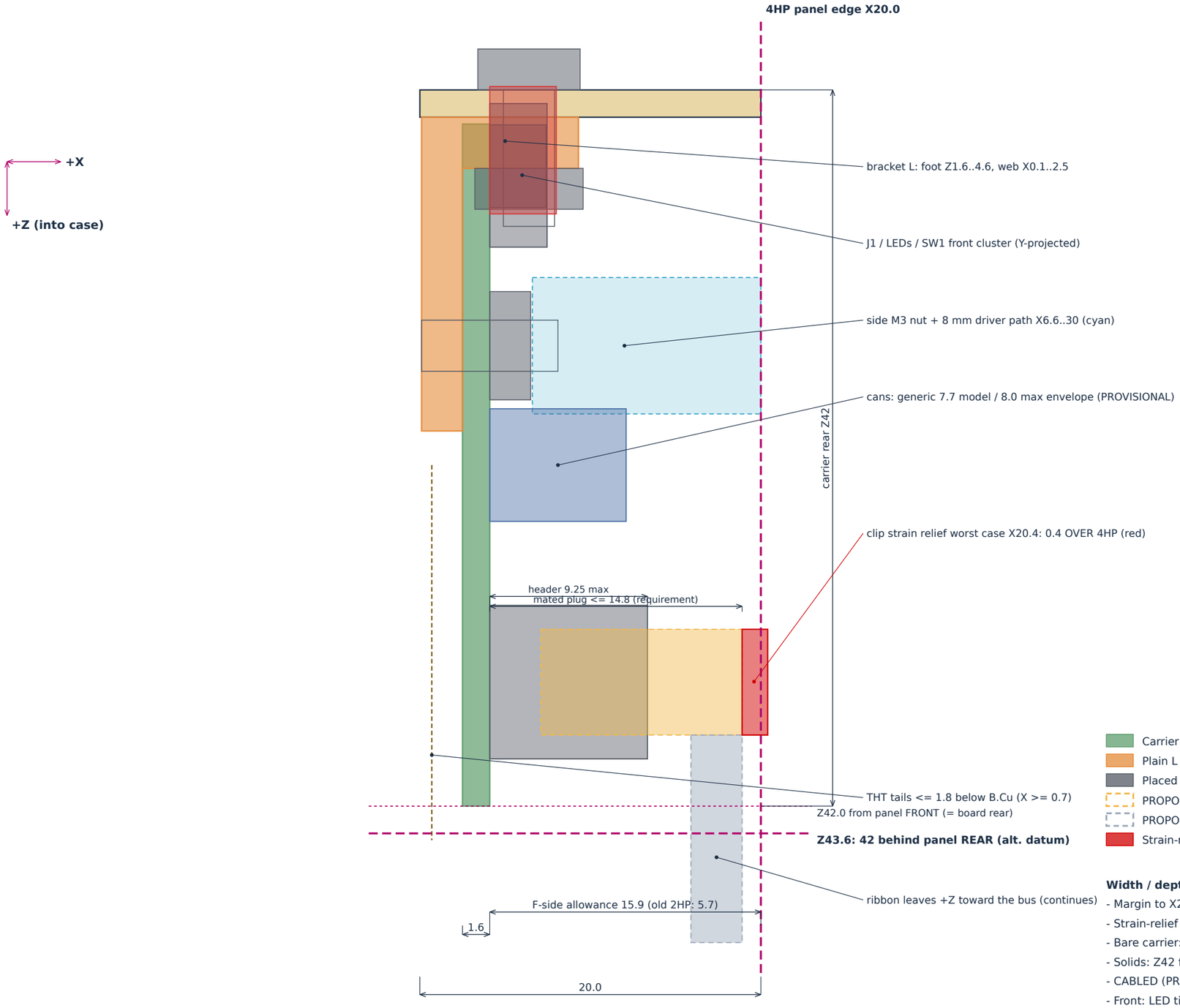
heuristic: boxes not unioned, cans double-counted; no fit/fail proof

B-side placement = later option only, not authorized

- M3: foot/web/nuts/driver paths clear of PCB and parts (nominal)

usb-power-45w - TOP VIEW / XZ STACK (seen from above, Y projected)

X across to the right, Z into the case downward. Heights of every placed part, brackets, M3 hardware and proposal envelopes against the 4HP width (X20.0) and the depth target.



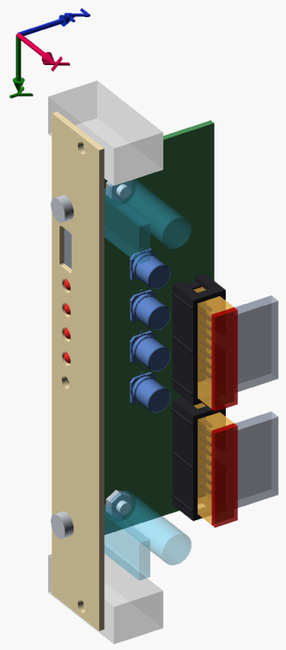
Carrier B.Cu X2.5 / F.Cu X4.1
Plain L bracket (copied)
Placed parts / M3 nuts / heads
PROPOSAL IDC plug
PROPOSAL ribbon
Strain-relief worst-case conflict

Width / depth summary (nominal):

- Margin to X20.0: header 6.65, plug 1.1, ribbon 1.1, cans 7.9, LEDs 12.0 mm
- Strain-relief clip worst case: -0.4 mm (CONFLICT unless measured lower)
- Bare carrier: 40.4 mm behind panel rear (Z42)
- Solids: Z42 from front = 40.4 behind rear
- CABLED (PROPOSAL envelope): Z50 from front = 48.4 behind rear -> NOT <= 42
- Front: LED tips 0.2 mm proud; M3 pan heads 2.4 mm proud of the panel front

usb-power-45w M1 - 3D assembly, front-right (F side)

Actual common-frame geometry: native panel + carrier STEP, brackets, M3 hardware, envelopes.
X across 4HP (20.0), Y down, Z into the case. Carrier perpendicular at B.Cu X2.5 / F.Cu X4.1.
Unrouted. Only J1, SW1, 4 LEDs, J2/J3 and 4 provisional cans are on the board.



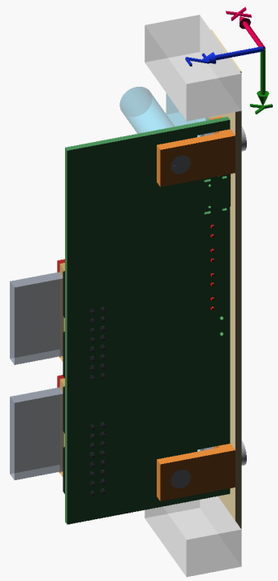
Legend (common assembly frame)

- Native 4HP panel PCB (kicad-cli STEP)
- Native carrier PCB, 100.5 x 40 x 1.6 (unrouted; parked SMD not shown)
- Two plain M3 L brackets (copied)
- M3x8 screws / M3 nuts (proxies)
- J2/J3 2x8 box headers (THT, KiCad model)
- D8/D16/D17/D18 3 mm LEDs (THT)
- C5/C34/C35/C36 cans (SMT) PROVISIONAL height context
- J1 USB-C / SW1 reset (SMT): drawing envelopes, no model
- PROPOSAL: mated IDC plug envelope
- PROPOSAL: ribbon leaving +Z to bus
- CONFLICT: clip strain relief worst case X20.4 > 20.0
- Tool paths: M3 nut driver / pliers
- Case rail section (context)

M1 MECHANICAL PREVIEW - UNROUTED - NOT A RELEASE - nominal mm

usb-power-45w M1 - 3D assembly, rear-left (B side, brackets)

Both L brackets on the carrier B face; M3 side screws from -X (flush heads), nuts on F.Cu.
Panel M3 screws from the front, nuts on the bracket foot. Ribbon envelopes leave toward +Z.
Rails shown translucent for context only.



Legend (common assembly frame)

- Native 4HP panel PCB (kicad-cli STEP)
- Native carrier PCB, 100.5 x 40 x 1.6 (unrouted; parked SMD not shown)
- Two plain M3 L brackets (copied)
- M3x8 screws / M3 nuts (proxies)
- J2/J3 2x8 box headers (THT, KiCad model)
- D8/D16/D17/D18 3 mm LEDs (THT)
- C5/C34/C35/C36 cans (SMT) PROVISIONAL height context
- J1 USB-C / SW1 reset (SMT): drawing envelopes, no model
- PROPOSAL: mated IDC plug envelope
- PROPOSAL: ribbon leaving +Z to bus
- CONFLICT: clip strain relief worst case X20.4 > 20.0
- Tool paths: M3 nut driver / pliers
- Case rail section (context)

M1 MECHANICAL PREVIEW - UNROUTED - NOT A RELEASE - nominal mm